

FinPilot AI - Complete Project Specification

Vision

FinPilot AI is an AI-powered personal finance and budgeting platform that not only tracks expenses but actively helps users save money through behavioral insights, gamification, predictive analytics, and intelligent recommendations.

Tech Stack

Frontend: HTML5, CSS3, Vanilla JavaScript

Backend: Node.js + Express.js

Database: MySQL

Charts: Chart.js

Authentication: JWT

Deployment: Frontend (Vercel/Netlify), Backend (Render/Railway), Managed MySQL.

Core Modules

- Authentication (Register/Login/Forgot Password)
- Dashboard
- Income Management
- Expense Management
- Budget Planner
- Analytics & Charts
- Calendar View
- Receipt Upload
- Notifications
- Reports (PDF/CSV)
- Admin Dashboard
- Profile & Settings

Unique Features

1. Should I Buy It? Advisor (affordability + goal impact)
2. Regret Predictor based on past purchases
3. Subscription Killer (unused subscriptions)
4. Daily AI Savings Challenges
5. Gamified Levels, XP & Badges
6. OCR Receipt Scanner
7. Personalized AI Financial Coach
8. Price Memory & Price Drop Alerts
9. Impulse Purchase Countdown Lock
10. Visual Dream Savings Goals
11. Weekly Funny Spending Roast
12. No-Spend Calendar & Streaks
13. Opportunity Cost Calculator
14. Future Savings Simulator
15. Hidden Expense Detector
16. Carbon Footprint per Purchase
17. Spending Heatmap by Location
18. Random Daily Saving Missions
19. Mood vs Spending Analysis
20. Accountability Buddy & Shared Goals
21. Financial Time Machine (show what savings become over months/years).

Database Tables

Users, Income, Expenses, Categories, Budgets, SavingsGoals, Receipts, Notifications, Reports, UserSettings, LoginHistory, Subscriptions, Achievements, Challenges, MoodLogs, PriceWatch, SharedGoals.

UI Pages

Landing, Login, Register, Dashboard, Income, Expenses, Analytics, Budget Planner, Calendar, Savings Goals, Reports, Notifications, Settings, Profile, Admin Dashboard.

Design

Modern glassmorphism UI, responsive layout, dark/light mode, animated charts, smooth transitions, progress rings, interactive cards, beautiful dashboards.

Development Instructions for Antigravity CLI

Build incrementally. First analyze the project and create the folder structure. Use MVC architecture. Design the MySQL schema first, then implement authentication, REST APIs, frontend pages, charts, AI insight engine (rule-based initially), and finally polish the UI. Keep code modular, documented, secure, and deployment-ready. Always explain the plan before making changes.